

8(a)(i)	number of protons: equal/same	B1
	number of neutrons: unequal/different	B1
8(a)(ii)	down (quark) changes to up (quark) or up down down (quarks) change to up up down (quarks)	B1
8(a)(iii)	(electron) antineutrino	B1
8(b)	charm (quark charge) is $(+)2/3(e)$ or 2 charm (quark charges) is $(+)4/3(e)$ or bottom (quark charge) is $-1/3(e)$	C1
	charge = $+2/3(e) + 2/3(e) - 1/3(e)$ $= (+)1(e)$	A1